Assignment 2: Progressive Electricity Billing

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Calculate household electricity bills across several months using multiple consumption slabs, fixed charges, and rebates. Next, iterate through the months to compare usage and identify any unusual spikes in power consumption.

Algorithm

1. Create a list of tuples containing (Month, Units Consumed).
2. Iterate through the list using a `for` loop.
3. Use nested conditional `if-elif-else` statements to calculate costs across billing slabs (e.g., 0-100, 101-300, 301+).
4. Add base fixed charges and apply rebates if units are extremely low.
5. Use a second loop to compare the current month's usage to the previous month's usage.
6. Print a warning if usage jumps significantly.

Python Implementation

```
monthly_usage = [("Jan", 90), ("Feb", 120), ("Mar", 130), ("Apr", 320), ("May",
310)]
fixed_charge = 50.0

print("--- Monthly Bill Statements ---")
for record in monthly_usage:
    month, units = record
    bill_amount = fixed_charge

    if units <= 100:
        bill_amount += units * 2.50
        bill_amount -= 10.0 # Energy saver rebate
    elif units <= 300:
        bill_amount += (100 * 2.50) + ((units - 100) * 4.00)
    else:
        bill_amount += (100 * 2.50) + (200 * 4.00) + ((units - 300) * 6.50)
        bill_amount += bill_amount * 0.05 # 5% surcharge for high consumption

    print("Bill for " + month + ": Rs. " + str(bill_amount))

print("\n--- Consumption Anomaly Check ---")
for i in range(1, len(monthly_usage)):
    prev_month, prev_units = monthly_usage[i-1]
    curr_month, curr_units = monthly_usage[i]

    diff = curr_units - prev_units
    if diff > 150:
        print("WARNING: Unusual usage spike detected in " + curr_month + " (+" +
str(diff) + " units).")
```